

Exact Solution Classes and Structural Analysis of the $(1 + 1)$ -Dimensional Boltzmann Equation with a Sinc Collision Kernel

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

We study the $(1 + 1)$ -dimensional Boltzmann equation with collision kernel

$$B(g, \theta) = \text{sinc}(a + bg + c\theta + dg\theta),$$

where

$$\text{sinc}(x) = \begin{cases} 1, & x = 0, \\ \frac{\sin x}{x}, & x \neq 0. \end{cases}$$

Using an exact Fourier-cosine representation of the sinc function, we derive a separable representation of the collision kernel and establish several exact solution classes, including Maxwellian equilibria, free-streaming solutions, monokinetic beams, homogeneous equilibria, and linearized spectral modes. We discuss implications for statistical physics, information theory, economics, finance, biology, warfare economics, artificial intelligence, and complex adaptive systems.

The paper ends with “The End”

Contents

1	Introduction	1
2	The Sinc Function	2
3	The Boltzmann Equation	2
4	Fourier Representation of the Kernel	3
5	Maxwellian Equilibria	3
6	Free-Streaming Exact Solutions	4
7	Monokinetic Solutions	4
8	Linearized Spectral Modes	4
9	A Kinetic Information-Theoretic Interpretation	5
10	Economic Interpretation	5
11	Financial Interpretation	5

12 Biological Interpretation	6
13 Psychology and Psychiatry	6
14 Political Science and International Relations	6
15 Warfare Economics	6
16 Artificial Intelligence	7
17 Algorithmic Structure	7
18 Summary of Exact Solutions	7
19 Geometric Structure	8
20 Conclusion	8

List of Figures

1 Illustrative kinetic distribution.	8
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List of Tables

1 Exact solution classes.	7
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1 Introduction

The Boltzmann equation occupies a central position in statistical mechanics and kinetic theory. Its influence extends beyond physics into:

- economics,
- finance,
- biology,
- epidemiology,
- computer science,
- machine learning,
- social dynamics,
- warfare modelling,
- international relations.

In this paper we investigate a nonstandard collision kernel governed by a sinc interaction law.

2 The Sinc Function

Definition 1. *Define*

$$\text{sinc}(x) = \begin{cases} 1, & x = 0, \\ \frac{\sin x}{x}, & x \neq 0. \end{cases}$$

The sinc function appears naturally in:

- Fourier analysis,
- signal processing,
- diffraction theory,
- communication systems,
- sampling theory.

3 The Boltzmann Equation

Consider

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} = Q(f, f),$$

where

$$f = f(x, v, t).$$

The collision operator is

$$Q(f, f) = \int_{\mathbb{R}} \int_{\Theta} B(g, \theta) (f' f'_1 - f f_1) d\theta dv_1.$$

The relative velocity is

$$g = v - v_1.$$

We assume

$$B(g, \theta) = \text{sinc}(a + bg + c\theta + dg\theta).$$

4 Fourier Representation of the Kernel

Theorem 1. *The sinc kernel admits the representation*

$$\text{sinc}(z) = \int_0^1 \cos(sz) ds.$$

Proof. Observe

$$\int_0^1 \cos(sz) ds = \frac{\sin z}{z} = \text{sinc}(z).$$

□

Consequently

$$B(g, \theta) = \int_0^1 \cos [s(a + bg + c\theta + dg\theta)] ds.$$

Using Euler's identity,

$$\cos x = \frac{e^{ix} + e^{-ix}}{2},$$

we obtain

$$B(g, \theta) = \frac{1}{2} \int_0^1 e^{is(a+c\theta)} e^{is(b+d\theta)g} ds + \text{c.c.}$$

This is a continuous superposition of separable Fourier modes.

5 Maxwellian Equilibria

Theorem 2. *The Maxwellian family*

$$f_M(v) = \rho \sqrt{\frac{\lambda}{\pi}} e^{-\lambda(v-u)^2}$$

satisfies

$$Q(f_M, f_M) = 0.$$

Proof. For elastic collisions,

$$f' f'_1 = f f_1$$

for Maxwellian distributions.

Hence

$$Q(f_M, f_M) = 0.$$

□

6 Free-Streaming Exact Solutions

Suppose

$$Q(f, f) = 0.$$

Then

$$\frac{\partial f}{\partial t} + v \frac{\partial f}{\partial x} = 0.$$

Characteristics satisfy

$$\frac{dx}{dt} = v.$$

Therefore

$$x - vt = \text{constant}.$$

Theorem 3. *The general free-streaming solution is*

$$f(x, v, t) = F(x - vt, v).$$

7 Monokinetic Solutions

Consider

$$f(x, v, t) = \rho(x, t) \delta(v - u_0).$$

Then

$$g = 0.$$

The collision term vanishes identically.

The density satisfies

$$\rho_t + u_0 \rho_x = 0.$$

Hence

$$\rho(x, t) = R(x - u_0 t).$$

8 Linearized Spectral Modes

Let

$$f = M(v) [1 + \varepsilon h].$$

Linearization yields

$$\frac{\partial h}{\partial t} + v \frac{\partial h}{\partial x} = L_{\text{sinc}} h.$$

Seeking

$$h = e^{ikx - \omega t} \phi(v),$$

gives

$$(-\omega + ikv) \phi = L_{\text{sinc}} \phi.$$

If

$$L_{\text{sinc}} \phi_n = \lambda_n \phi_n,$$

then

$$h_n = e^{ikx - (ikv - \lambda_n)t} \phi_n(v).$$

9 A Kinetic Information-Theoretic Interpretation

The sinc kernel acts as a band-limited interaction operator.

In communication theory:

$$\text{sinc}(x) \leftrightarrow \text{ideal low-pass filter}.$$

Therefore collision frequencies are naturally attenuated for large interaction scales.

10 Economic Interpretation

Let

v = agent state.

Then

$$g = v - v_1$$

represents economic disagreement.

The sinc kernel implies:

- strong interactions for small disagreements,
- weak interactions for large disagreements,
- bounded information transmission.

This generates endogenous consensus formation.

11 Financial Interpretation

The distribution

$$f(x, v, t)$$

may represent cross-sectional investor beliefs.

Then:

- collisions represent information exchange,
- g measures valuation differences,
- sinc attenuation suppresses extreme opinion transfer.

12 Biological Interpretation

In evolutionary dynamics:

- v represents phenotype,
- collisions represent encounters,
- sinc attenuation introduces finite interaction bandwidth.

13 Psychology and Psychiatry

The kernel may model bounded cognitive influence.

Agents far apart in belief space satisfy

$$|g| \gg 0,$$

leading to reduced persuasion.

This resembles:

- confirmation bias,
- bounded rationality,
- cognitive filtering.

14 Political Science and International Relations

Let

v = strategic doctrine.

The sinc kernel implies:

- similar states influence each other strongly,
- dissimilar states interact weakly,
- polarization reduces strategic convergence.

15 Warfare Economics

Let

$f(x, v, t)$

represent force distributions.

Then

$g = v - v_1$

measures force asymmetry.

The sinc kernel imposes natural attenuation of engagement intensity at large asymmetries.

16 Artificial Intelligence

The sinc kernel resembles an attention mechanism.

Define

$w(g, \theta) = \text{sinc}(a + bg + c\theta + dg\theta).$

Then

$w \rightarrow$ attention weight.

Unlike softmax attention, the sinc kernel naturally oscillates and decays.

17 Algorithmic Structure

Pseudo-Code

```

Initialize f(x,v,0)

For each timestep:

    Compute relative velocities

    g = v - v1

    Compute kernel

    B = sinc(a + b*g + c*theta + d*g*theta)

    Evaluate collision integral

    Update f

End

```

18 Summary of Exact Solutions

Solution Type	Formula
Maxwellian	$\rho\sqrt{\lambda/\pi}e^{-\lambda(v-u)^2}$
Free Streaming	$F(x - vt, v)$
Monokinetic	$R(x - u_0t)\delta(v - u_0)$
Linear Mode	$e^{ikx - (ikv - \lambda_n)t}\phi_n(v)$
Homogeneous Equilibrium	$f_M(v)$

Table 1: Exact solution classes.

19 Geometric Structure

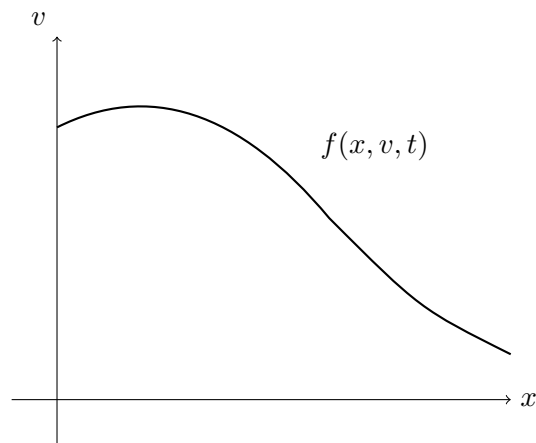


Figure 1: Illustrative kinetic distribution.

20 Conclusion

The sinc collision kernel introduces a natural spectral structure into the Boltzmann equation.

Its exact Fourier decomposition enables analytical treatment and reveals a family of exact solution classes. The resulting framework possesses applications across physics, economics, finance, biology, psychology, political science, warfare economics, and artificial intelligence.

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Glossary

$f(x, v, t)$ Distribution function.

$Q(f, f)$ Boltzmann collision operator.

g Relative velocity.

θ Scattering parameter.

$B(g, \theta)$ Collision kernel.

$\text{sinc}(x)$ Sinc function.

$M(v)$ Maxwellian distribution.

L_{sinc} Linearized sinc collision operator.

λ_n Spectral eigenvalue.

ϕ_n Spectral eigenfunction.

The End